

# PRINCESS CROAK'S PALACE · PRINCESS CROAK · EVERY UNIT, EVERY DIFFICULTY

What every gnome, fairy, wasp, frog and Princess Croak herself does to your men on EASY, MEDIUM, HARD and MEGA, with every body's HP and head-count · read out of the live game v28.88 on Nov 1, 2026 04:21 EST · every damage number checked on the game's own machines (Chrome 244/244 · WebKit 244/244) · for Raziya

**The one thing to know:** in the palace **almost every hit takes BAR HITS off a man, not a KO**: her harp shot (**2 bar hits**, MEGA **3**), the guard's jab (1), the knight's jab (2), the slinger's glob (1), the wasp's sting (1, and the man is held 2 s), the hornet's sting (3, and he is bowled over), the frog's splat and tongue (1 each), and the sage's fire (1 a beat, ten beats). A level-70 rifleman has 7 bar hits (33 HP), so her harp KOs him on the **4th shot** on EASY to HARD and the **3rd on MEGA**. Three things are a KO whatever a man's HP: **the gnome guard's RAM** (a charge at a man ahead within 300 px; a jeep or the tank is destroyed on the spear; the saber turns it about half the time, THE RIOT SHIELD's 5 and the Claymore's counted block eat it), **her LIGHT ARROWS** (MEDIUM up, at half her bar; MEGA again at a quarter: every man a bolt crosses, through every save: the one hit nothing turns), **her FACE SHOT** (a Sarge within 84 px: through the saber's block; the riot shield and the Claymore's block eat it, the deflect dome turns it) — and the sage's toad curse takes Sarge out of the fight for 5 s. Difficulty changes her bar (×2 · ×2.4 · ×6 · ×9), her lives (1 · 1 · 3 · 3), her cadence, the court's size and HP, and brings the arrows, the knights, the sages and the hornets.

THE BOSS LAB fights the palace at **LEVEL 70 · 77 · 83 · 90** (EASY · MEDIUM · HARD · MEGA): Sarge with the saber, the scout jeep, the dogfighter, the tank, the bomber and seven men at that rank (rifleman, M. gunner, bazooka, mortar, medic, flame, sweeper). The radio man, grenadier, commando and spy are a tap away in THE PLATOON. Sarge wears the two rank-5 gifts (telekinesis on the saber; the riot shield answers only with the shotgun in hand). Every "level" number on this page is that army.

## What changed recently (the palace's own cards)

| live since | when | card | what it does in this fight |
|---|---|---|---|
| v28.74 | Oct 31 | **CARD-HP3 · THE HEAVIES** | +3 HP every tenth rank for the bazooka, grenadier, flame and commando. At level 70: bazooka 66, grenadier 61, flame 80, commando 75 HP. Every palace hit counts in bar hits, so more HP means each bar hit is worth more HP, not fewer hits. |
| v28.68 | Oct 30 | **CARD-BRB · THE BOMBER** | Your bomber flies in every boss fight now, the palace included. |
| v28.65 | Oct 30 | **CARD-LRB · ONE HP NUMBER** | Your man's HP is the number on his unit sheet. A palace hit still counts in bar hits, but lands a little lighter on tough men: 90 % + 10 % × bar ÷ HP of its weight (a level-70 rifleman takes 1.84 of 2 from her harp, a flame 1.82). |
| v28.46 | Oct 30 | **CARD-BLP · THE BOSS LAB'S STOCK ARMY** | The lab fights the palace with its own army at LEVEL 70 · 77 · 83 · 90 (EASY · MEDIUM · HARD · MEGA): the one this page reads. |
| card 1213 | — | **CARD-LA2 · THE KNIGHT'S MIRROR** | A knight's shield met from the front by a Sarge's own light arrow (the sniper mod won on HARD) throws it back at the shooter. |
| card 1027 | — | **CARD-ML-A3 · PRINCESS CROAK, THE SIXTH HERO** | Beat her on MEGA and she joins as a hero (her HP row sits among the heroes above). |
| card 1023 | Oct 23 | **CARD-PCS · PALACE SPACING** | Laura's spacing pass on her palace picture (the look; no fight numbers). |
| 10/10 | Oct 10 | **CARD-PC2 · THE BIGGER COURT** | Wyatt: "give her more troops coming in" and "There should never be two Princess Croaks out at once": HARD and MEGA field more guards, slingers, wasps, knights, frogs; two hornets on HARD, three on MEGA; three sages at MEGA's open; THE COURT RETURNS ×8 on HARD and MEGA; one princess at every seat count. |
| card 1014 | — | **THE PRESSURE · THE PALACE TUNE** | Her harp shot per tier (the cadence 2.4 · 2.0 · 1.8 · 2.0 s, MEGA 3 bar hits), the gate clocks, the knights at 75 %, the guard turning out at 50 % and 30 %, THE COURT RETURNS on the clock, the court's HP cells. |

| live since | when | card | what it does in this fight |
|---|---|---|---|
| CARD-P3 | — | **THE RETREAT · THE BUBBLE COURT** | HARD and MEGA: three bars; at a zero with bars left she flies back to her stage and calls her fairies and guards; the bar clock; the big guards charge at ×2.2. |
| card 1032 | Sep 14–15 | **CARD-PM-A · THE THRONE ROOM** | Wyatt: "Princess croaks map also needs to be redesigned to feel more epic". Six marble columns as cover, two frog fountains, the water stairs as a river with two bridges, the pool, her throne and harp behind her top step; MEGA moonlit. |
| card 0947e | Sep 11 | **THE PALACE RAID** | Wyatt: "at half HP point, is going to begin to charge her light arrow… three large light arrows… if a commando has his reflect, you may reflect it right back at her… built-in auto-kill mechanics for any Sarge that comes too close". The fight as designed. |
| — | — | **NOT ON THIS MAC** | The Big Mac's git log is not on Maker 1 (this worktree is a shallow clone at the Big Mac's main 04d094bb69). The rows above are read off the client's own comments and the other kits' records; a version is given where one is known. |

## How a hit is counted

Your men carry **HP**: the number on their unit sheet (a level-70 flame has 80, a rifleman 33). Their health bar is also cut into a few **bar hits** (7 for most men at level 70–90, 6 for the medic and spy, 8 for the commando, 12 for the saber Sarge, 23 for the Claymore). Three kinds of hit in this fight:

- **Bar hits** (everything the court does, and her harp): a hit of N bar hits lands 90 % + 10 % × bar ÷ HP of N since v28.65, so a level-70 rifleman takes 1.84 of his 7 from her harp (8.7 HP of 33) and 0.92 from a guard's jab. Bar hits mend after 4 s without a hit. A hero's bar IS his HP pool (a bar hit lands whole on it).
- **KO hits** (the guard's ram, her light arrows, her face shot): out in one hit whatever his HP. Only the arrows go through every save. Her face shot goes through the saber's block (the shot is sent with the saber's roll switched off: the KO door's rolled flag, by design, the auto-kill for a Sarge who comes too close), but THE RIOT SHIELD (5 a battle, the shotgun) and the Claymore's counted block (1 a battle: THE DUELIST, every Claymore's default spec; three only with FEAR or WALL picked) eat it whole, and the deflect dome turns it: measured, the shotgun Sarge fell to the 6th face shot and the Claymore to the 2nd. The ram reaches that door with no roll spent, so every save answers it: the saber turns it about half the time (measured 6/10 · 7/8 · 6/3 · 9/11 of 16 fresh Sarges KO'd on E · M · H · G, Chrome/WebKit), the plate and the counted block eat it (the shotgun Sarge fell to the 6th ram, the Claymore to the 2nd), the pet shield, the blue wall and the teal turn it; the deflect dome does not (its KO door catches only the officer's potshot and her face shot).
- **No hit**: the fairy mends her court and bubbles a guard BIG; the bubble itself pops on a green round; the sage's curse turns Sarge into a toad for 5 s (no fire, no swing).

Saves still roll on every bar hit: a jetting, dropping or dragged man cannot be touched, the pet shield, the deflect dome, **the saber Sarge's block turns a bar hit 50 % of the time** (measured: 6 of 15, 17 of 29, 4 of 11, 3 of 8 of her shots on the four tiers), **the shotgun Sarge's riot shield eats the first 5 hits whole** (measured), and the Claymore's counted block eats 1 (measured; THE DUELIST's one block is every Claymore's default spec, the table's 3 only with FEAR or WALL picked). The same saves answer the guard's ram, all but the deflect dome (its KO door catches only the officer's potshot and her face shot); her face shot skips only the saber's roll; her light arrows skip them all. A charged commando catches a light arrow and sends it back at her.

## The strikes · what a hit does by difficulty

| hit | EASY | MEDIUM | HARD | MEGA | who it hits | since |
|---|---|---|---|---|---|---|
| **HER HARP SHOT · bar hits** | 2 | 2 | 2 | 3 | the nearest man within 400 px, every 2.4 · 2 · 1.8 · 2 s (a level-70 rifleman: 8.7 HP a shot on EASY, 16.1 on MEGA) | the per-tier cadence and MEGA's third bar hit: card 1014 |
| **HER FACE SHOT** | KO | KO | KO | KO | a Sarge within 84 px: the hop back and a shot through his saber block (THE RIOT SHIELD's 5 and the Claymore's counted block eat it; the deflect dome turns it) | unchanged |
| **HER LIGHT ARROWS** | — | KO | KO | KO ×2 | every man a bolt crosses (three lanes 120 px apart, 56 px either side); a charged commando, the blue wall or the teal catches one | MEGA's second volley: card 1014 |
| **THE GNOME'S RAM** | KO | KO | KO | KO | a man ahead within 300 px; a jeep or the tank is destroyed on the spear; the saber turns it about half the time, THE RIOT SHIELD's 5 and the Claymore's counted block eat it, the pet shield, the blue wall and the teal turn it (the deflect dome does not) | unchanged |
| **THE GNOME'S JAB · bar hits** | 1 | 1 | 1 | 1 | a man within 28 px | unchanged |
| **THE KNIGHT'S JAB · bar hits** | — | 2 | 2 | 2 | a man within 28 px | unchanged |
| **THE SLINGER'S GLOB · bar hits** | 1 | 1 | 1 | 1 | a man within 280 px | unchanged |
| **THE SAGE'S FIRE · bar hits a beat** | — | 1 | 1 | 1 | your three rearmost men, 10 beats over 5 s | unchanged |
| **THE SAGE'S TOAD CURSE** | — | 5 s a toad | 5 s a toad | 5 s a toad | a Sarge within 200 px: no fire, no swing, hops at ×0.6 | unchanged |
| **THE WASP'S STING · bar hits** | 1 | 1 | 1 | 1 | every man a dive crosses; held 2 s | unchanged |
| **THE HORNET'S STING · bar hits** | — | — | 3 | 3 | every man a dive crosses; bowled over | two hornets on HARD, three on MEGA: CARD-PC2 |
| **THE FROG'S SPLAT · TONGUE · bar hits** | 1 · 1 | 1 · 1 | 1 · 1 | 1 · 1 | a man within 150 px (the hop) · 56 px (the tongue) | unchanged |
| **THE FAIRY'S BUBBLE** | BIG gnome: ×4 HP | BIG gnome: ×4 HP | BIG gnome: ×4 HP | BIG gnome: ×4 HP | her own court, never your men | unchanged |

## Who survives · the lab's level

How many of that hit in a row (inside the 4 s mend) KO a fresh man at the lab's level for that tier (EASY 70 · MEDIUM 77 · HARD 83 · MEGA 90). **KO** = one hit. Hover a cell for the numbers.

KO one hit · 2 two · 3 three · 4 four · 5+ five or more

· a dot in the corner = the count measured on the game's own machines in this read (both engines agree) · +½ = the saber turns about half (her harp shot, the ram) · +5 = the riot shield eats 5 first · +1 on the Claymore = his counted block eats 1 first · "·" = not in this fight / only a Sarge

| man | HP<br>E · M · H · G | bar hits | HER HARP SHOT | | | | GNOME'S JAB | | | | KNIGHT'S JAB | | | | SLINGER'S GLOB | | | | WASP'S STING | | | | HORNET'S STING | | | | FROG'S SPLAT · TONGUE | | | | SAGE'S FIRE (beats) | | | | GNOME RAM | LIGHT ARROWS | | | | FACE SHOT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | E | M | H | G | E | M | H | G | E | M | H | G | E | M | H | G | E | M | H | G | E | M | H | G | E | M | H | G | E | M | H | G | | E | M | H | G | |
| THE LAB'S STOCK ARMY (LEVEL 70 · 77 · 83 · 90) | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| RIFLEMAN | 33 · 33 · 37 · 41 | 7 | 4 | 4 | 4 | 3 | 8 | 8 | 8 | 8 | · | 4 | 4 | 4 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | · | · | 3 | 3 | 8 | 8 | 8 | 8 | · | 8 | 8 | 8 | KO | · | KO | KO | KO | · |
| M. GUNNER | 50 · 50 · 55 · 60 | 7 | 4 | 4 | 4 | 3 | 8 | 8 | 8 | 8 | · | 4 | 4 | 4 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | · | · | 3 | 3 | 8 | 8 | 8 | 8 | · | 8 | 8 | 8 | KO | · | KO | KO | KO | · |
| BAZOOKA | 66 · 66 · 75 · 83 | 7 | 4 | 4 | 4 | 3 | 8 | 8 | 8 | 8 | · | 4 | 4 | 4 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | · | · | 3 | 3 | 8 | 8 | 8 | 8 | · | 8 | 8 | 8 | KO | · | KO | KO | KO | · |
| MORTAR | 30 · 30 · 33 · 37 | 7 | 4 | 4 | 4 | 3 | 8 | 8 | 8 | 8 | · | 4 | 4 | 4 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | · | · | 3 | 3 | 8 | 8 | 8 | 8 | · | 8 | 8 | 8 | KO | · | KO | KO | KO | · |
| MEDIC | 29 · 29 · 33 · 36 | 6 | 4 | 4 | 4 | 3 | 7 | 7 | 7 | 7 | · | 4 | 4 | 4 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | · | · | 3 | 3 | 7 | 7 | 7 | 7 | · | 7 | 7 | 7 | KO | · | KO | KO | KO | · |
| FLAME | 80 · 80 · 90 · 100 | 7 | 4 | 4 | 4 | 3 | 8 | 8 | 8 | 8 | · | 4 | 4 | 4 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | · | · | 3 | 3 | 8 | 8 | 8 | 8 | · | 8 | 8 | 8 | KO | · | KO | KO | KO | · |
| SWEEPER | 31 · 31 · 35 · 38 | 7 | 4 | 4 | 4 | 3 | 8 | 8 | 8 | 8 | · | 4 | 4 | 4 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | · | · | 3 | 3 | 8 | 8 | 8 | 8 | · | 8 | 8 | 8 | KO | · | KO | KO | KO | · |
| SARGE · SABER | 40 · 41 · 44 · 47 | 12 | 7+½ | 7+½ | 7+½ | 5+½ | 13 | 13 | 13 | 13 | · | 7 | 7 | 7 | 13 | 13 | 13 | 13 | 13 | 13 | 13 | 13 | · | · | 5 | 5 | 13 | 13 | 13 | 13 | · | 13 | 13 | 13 | KO+½ | · | KO | KO | KO | KO |
| A TAP AWAY (THE PLATOON, SARGE'S OTHER IRONS) | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| RADIO MAN | 31 · 31 · 35 · 38 | 7 | 4 | 4 | 4 | 3 | 8 | 8 | 8 | 8 | · | 4 | 4 | 4 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | · | · | 3 | 3 | 8 | 8 | 8 | 8 | · | 8 | 8 | 8 | KO | · | KO | KO | KO | · |
| GRENADIER | 61 · 61 · 68 · 76 | 7 | 4 | 4 | 4 | 3 | 8 | 8 | 8 | 8 | · | 4 | 4 | 4 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | · | · | 3 | 3 | 8 | 8 | 8 | 8 | · | 8 | 8 | 8 | KO | · | KO | KO | KO | · |
| COMMANDO | 75 · 75 · 84 · 93 | 8 | 5 | 5 | 5 | 3 | 9 | 9 | 9 | 9 | · | 5 | 5 | 5 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | · | · | 3 | 3 | 9 | 9 | 9 | 9 | · | 9 | 9 | 9 | KO | · | KO | KO | KO | · |
| SPY | 36 · 36 · 41 · 45 | 6 | 4 | 4 | 4 | 3 | 7 | 7 | 7 | 7 | · | 4 | 4 | 4 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | · | · | 3 | 3 | 7 | 7 | 7 | 7 | · | 7 | 7 | 7 | KO | · | KO | KO | KO | · |
| SARGE · PISTOL | 41 · 41 · 44 · 47 | 11 | 6 | 6 | 6 | 4 | 12 | 12 | 12 | 12 | · | 6 | 6 | 6 | 12 | 12 | 12 | 12 | 12 | 12 | 12 | 12 | · | · | 4 | 4 | 12 | 12 | 12 | 12 | · | 12 | 12 | 12 | KO | · | KO | KO | KO | KO |
| SARGE · SHOTGUN | 40 · 41 · 44 · 47 | 8 | 5+5 | 5+5 | 5+5 | 3+5 | 9 | 9 | 9 | 9 | · | 5 | 5 | 5 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | · | · | 3 | 3 | 9 | 9 | 9 | 9 | · | 9 | 9 | 9 | KO+5 | · | KO | KO | KO | KO+5 |
| SARGE · SNIPER | 31 · 31 · 34 · 37 | 8 | 5 | 5 | 5 | 3 | 9 | 9 | 9 | 9 | · | 5 | 5 | 5 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | · | · | 3 | 3 | 9 | 9 | 9 | 9 | · | 9 | 9 | 9 | KO | · | KO | KO | KO | KO |
| SARGE · CLAYMORE | 71 · 73 · 73 · 76 | 23 | 13+1 | 13+1 | 13+1 | 9+1 | 25 | 25 | 25 | 25 | · | 13 | 13 | 13 | 25 | 25 | 25 | 25 | 25 | 25 | 25 | 25 | · | · | 9 | 9 | 25 | 25 | 25 | 25 | · | 25 | 25 | 25 | KO+1 | · | KO | KO | KO | KO+1 |

The bar is 7 for most men from level 70 to 90, so the counts hold across the tiers; HP is what each bar hit is worth. HER HARP SHOT: every man and iron measured on every tier (the dots). The court's hits were measured on a rifleman (jab, glob, sting, frog, fire), a flame (knight, hornet, fire) and a grenadier (fire); the rest of the column is the same arithmetic. SAGE'S FIRE counts beats (10 in a ring's 5 s life: a man who stands in it is KO'd). GNOME RAM, LIGHT ARROWS and FACE SHOT: one hit, whatever the HP; only the arrows pass every save (the ram was measured on a rifleman, a flame and Sarge's saber, pistol, shotgun and Claymore; the face shot on all four irons). SARGE · SABER's harp cells carry no dot: 7 · 7 · 7 · 5 is the count if every landed shot came inside the 4 s mend, but his block spaces them and his bar mends between, so the machines measured 9/11 · 12/9 · 7/7 · 5/7 landed shots to his KO (E · M · H · G, Chrome/WebKit).

## The heroes at the lab's level (one in command), measured

| hero | HP (E · M · H · G) | bar hits | her harp shot | shots to KO | measured: shots landed in 60 s (Chrome/WebKit where they differ) | her face shot | her light arrow |
|---|---|---|---|---|---|---|---|
| KING YETI | **138 · 139 · 176 · 213** | 126 · 126 · 182 · 238 | 2 · 2 · 2 · 3 of his bar a shot | 63 · 63 · 91 · 80 | 24 · 29 · 32 · 29 | never: not a Sarge | KO by the code (the same door as a man's); not staged here |
| KING REX | **209 · 211 · 231 · 239** | 137 · 137 · 163 · 169 | 2 · 2 · 2 · 3 of his bar a shot | 69 · 69 · 82 · 57 | 25 · 29 · 33 · 29 | never: not a Sarge | KO by the code (the same door as a man's); not staged here |
| KING CROAK | **117 · 119 · 126 · 133** | 77 · 77 · 83 · 89 | 2 · 2 · 2 · 3 of his bar a shot | 39 · 39 · 42 · 30 | 25 · 25 · 29 · 26 | never: not a Sarge | KO by the code (the same door as a man's); not staged here |
| JANE | **126 · 128 · 135 · 142** | 68 · 68 · 74 · 80 | 2 · 2 · 2 · 3 of his bar a shot | 34 · 34 · 37 · 27 | 25 · 29 · 33 · 27 KO | never: not a Sarge | KO by the code (the same door as a man's); not staged here |
| EMPEROR BEAR | **218 · 221 · 241 · 249** | 137 · 137 · 163 · 169 | 2 · 2 · 2 · 3 of his bar a shot | 69 · 69 · 82 · 57 | 20/22 · 23/25 · 29 · 21/16 | never: not a Sarge | KO by the code (the same door as a man's); not staged here |
| PRINCESS CROAK | **97 · 98 · 104 · 111** | 77 · 77 · 83 · 89 | 2 · 2 · 2 · 3 of his bar a shot | 39 · 39 · 42 · 30 | 20/21 · 25/24 · 29 · 25 | never: not a Sarge (measured at 60 px) | KO by the code (the same door as a man's); not staged here |

A hero's bar is his HP pool: her harp takes 2 (MEGA 3) of it a shot, whole (measured on every tier). Only Jane fell inside 60 s (MEGA: 27 shots on both engines). A hero walks, so the count landed in 60 s is the engines' clocks as much as hers: EMPEROR BEAR on EASY (20 on Chrome, 22 on WebKit); EMPEROR BEAR on MEDIUM (23 on Chrome, 25 on WebKit); EMPEROR BEAR on MEGA (21 on Chrome, 16 on WebKit); PRINCESS CROAK on EASY (20 on Chrome, 21 on WebKit); PRINCESS CROAK on MEDIUM (25 on Chrome, 24 on WebKit); every shot's weight was the same on both. Her face shot is for a Sarge alone (measured on the Princess Croak hero). The guard's ram and her light arrows go through the same doors for a hero as for a man (the code; not staged on this page).

## The level a man first survives

For her harp shot and the hornet's sting (the hits that take part of a man). The KO hits have no such level: no rank survives them. **Green** = he already does at the lab's EASY level 70; — = not even at rank 100.

| | | | HER SHOTS IN A ROW · EASY–HARD (2 bar hits) | | | | … MEGA (3) | | | | HORNET STINGS | | HER SHOTS TO KO · EASY–HARD, BY RANK | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| man | HP / bar at 0 | HP / bar at 100 | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 | 1 | 2 | r0 | r10 | r25 | r50 | r70 | r90 | r100 |
| RIFLEMAN | 3 / 3 | 45 / 8 | 0 | 5 | 19 | 100 | 2 | 19 | — | — | 2 | 19 | 2 | 3 | 4 | 4 | 4 | 4 | 5 |
| M. GUNNER | 8 / 3 | 65 / 8 | 0 | 5 | 19 | 100 | 0 | 19 | — | — | 0 | 19 | 2 | 3 | 4 | 4 | 4 | 4 | 5 |
| BAZOOKA | 3 / 3 | 92 / 8 | 0 | 6 | 19 | 100 | 2 | 19 | — | — | 2 | 19 | 2 | 3 | 4 | 4 | 4 | 4 | 5 |
| MORTAR | 3 / 3 | 40 / 8 | 0 | 6 | 19 | 100 | 2 | 19 | — | — | 2 | 19 | 2 | 3 | 4 | 4 | 4 | 4 | 5 |
| MEDIC | 2 / 2 | 40 / 7 | 3 | 10 | 42 | — | 8 | 42 | — | — | 8 | 42 | 1 | 3 | 3 | 4 | 4 | 4 | 4 |
| FLAME | 6 / 3 | 110 / 8 | 0 | 5 | 19 | 100 | 0 | 19 | — | — | 0 | 19 | 2 | 3 | 4 | 4 | 4 | 4 | 5 |
| SWEEPER | 3 / 3 | 42 / 8 | 0 | 6 | 19 | 100 | 2 | 19 | — | — | 2 | 19 | 2 | 3 | 4 | 4 | 4 | 4 | 5 |
| RADIO MAN | 3 / 3 | 42 / 8 | 0 | 6 | 19 | 100 | 2 | 19 | — | — | 2 | 19 | 2 | 3 | 4 | 4 | 4 | 4 | 5 |
| GRENADIER | 3 / 3 | 84 / 8 | 0 | 6 | 19 | 100 | 2 | 19 | — | — | 2 | 19 | 2 | 3 | 4 | 4 | 4 | 4 | 5 |
| COMMANDO | 7 / 4 | 102 / 9 | 0 | 0 | 10 | 42 | 0 | 10 | 100 | — | 0 | 10 | 3 | 4 | 4 | 5 | 5 | 5 | 5 |
| SPY | 2 / 2 | 50 / 7 | 3 | 10 | 42 | — | 8 | 42 | — | — | 8 | 42 | 1 | 3 | 3 | 4 | 4 | 4 | 4 |
| SARGE · SABER | 9 / 8 | 50 / 13 | 0 | 0 | 0 | 0 | 0 | 0 | 5 | 42 | 0 | 0 | 5 | 6 | 6 | 7 | 7 | 7 | 8 |

| | | | HER SHOTS IN A ROW · EASY–HARD (2 bar hits) | | | | … MEGA (3) | | | | HORNET STINGS | | HER SHOTS TO KO · EASY–HARD, BY RANK | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| man | HP / bar at 0 | HP / bar at 100 | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 | 1 | 2 | r0 | r10 | r25 | r50 | r70 | r90 | r100 |
| **SARGE · PISTOL** | 8 / 4 | 50 / 12 | 0 | 0 | 10 | 42 | 0 | 10 | 50 | 100 | 0 | 10 | 3 | 4 | 4 | 6 | 6 | 6 | 7 |
| **SARGE · SHOTGUN** | 9 / 4 | 50 / 9 | 0 | 0 | 10 | 42 | 0 | 10 | 100 | — | 0 | 10 | 3 | 4 | 4 | 5 | 5 | 5 | 5 |
| **SARGE · SNIPER** | 6 / 4 | 40 / 9 | 0 | 0 | 10 | 42 | 0 | 10 | 100 | — | 0 | 10 | 3 | 4 | 4 | 5 | 5 | 5 | 5 |
| **SARGE · CLAYMORE** | 22 / 19 | 78 / 24 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 10 | 12 | 12 | 13 | 13 | 13 | 13 |

Read from the game's own HP and bar ladders at every rank 0–100 (the tenth-rank extras included; no gifts, soul or gear). Sarge's rows read the iron at the same rank. HER SHOTS TO KO by rank on EASY to HARD (2 bar hits a shot); MEGA's third bar hit takes one shot off most men.

## The units

| unit | attack | what a hit does | HP EASY | MED | HARD | MEGA | n EASY | MED | HARD | MEGA | leak | tags |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
|  **PRINCESS CROAK (the boss)** walks 24 px/s, keeps 240 px off your line and backs off at 96 px/s inside 160 px · lives 1 · 1 · 3 · 3 | **HARP SHOT · FACE SHOT · LIGHT ARROWS** a harp shot at the nearest man within 400 px every 2.4 · 2 · 1.8 · 2 s · a Sarge within 84 px: the face shot · at half her bar (MEDIUM up) the three light arrows | **2 · 2 · 2 · 3 bar hits** a harp shot (E · M · H · G) · the face shot and the light arrows are a **KO** | 320 | 384 | 960 | 1440 | 1 | 1 | 1 | 1 | 8 | 400 |
|  **THE GNOME GUARD** walks 26 px/s · his shield turns every 2nd round from the front; bullets do half, blasts double · a fairy's bubble makes him BIG (×4 HP) · Princess Croak's summoned guards charge the line at ×2.2 once big | **RAM · JAB** braces 0.45 s then charges 380 px/s at a man ahead within 300 px, 100 px past him, then rests 1.6 s · jabs a man within 28 px every 1 s (0.2 s wind-up) | the ram a **KO** unless a save answers (the saber about half the time, THE RIOT SHIELD's 5, the Claymore's counted block; a jeep or the tank destroyed on the spear) · the jab **1 bar hit** | 8 | 10 | 12 | 14 | 20 | 27 | 72 +3 a call | 94 +4 a call | 1 | 18 |
|  **THE GNOME KNIGHT** MEDIUM up · walks 20 px/s · his shield turns EVERY round from the front (shoot him from behind or the flank); bullets do half, blasts double · a shield met by a light arrow from a Sarge's mod throws it back | **JAB** jabs a man within 28 px every 1.4 s (0.35 s wind-up); no ram | **2 bar hits** | — | 18 | 21 | 25 | — | 2 | 4 | 5 | 2 | 36 |

| unit | attack | what a hit does | HP EASY | MED | HARD | MEGA | n EASY | MED | HARD | MEGA | leak | tags |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
|  **THE GNOME SLINGER**<br>walks 24 px/s | **GLOB**<br>aims 0.6 s then slings at a man within 280 px, every 3 s and the aim (measured 3.2–3.6 · 3.1–3.5 · 3.1–3.4 · 3.2–3.6 s) · backs off from a man inside 120 px | **1 bar hit** | 4 | 5 | 6 | 7 | 2 | 3 | 6 | 7 | 1 | 12 |
|  **THE GNOME SAGE**<br>MEDIUM up (one at the pond; HARD three; MEGA three at the open and two at the pond) · walks 22 px/s | **FIRE RINGS · TOAD CURSE**<br>the first fire at 6 s then every 12 s: the staff up 0.6 s, the rings 1.0 s, then 3 patches 56 px around their centres (112 px across) · the curse on a Sarge within 200 px, a 0.6 s cast, every 15 s | the fire **1 bar hit a beat**, a beat every 0.5 s for 5 s (10 beats) under each of your three rearmost men · the curse turns **Sarge** into a toad for 5 s (he hops at ×0.6, cannot fire or swing) | — | 13 | 15 | 18 | — | 1 | 3 | 5 | 2 | 30 |
|  **THE FAIRY**<br>hovers 80 px behind the front man · Princess Croak's summoned fairies blow bubbles and never mend | **MEND · BUBBLE**<br>the first bubble at 4 s, then every 10 s (0.6 s blow, 120 px/s); one big man a fairy at a time | no hit on your men: mends her court **1 HP a second** within 120 px and blows a bubble that makes the front gnome **BIG: ×4 HP, ×1.5 size, jabs ×1.5 as often** | 3 | 4 | 5 | 5 | 1 | 2 | 2 +3 a call | 2 +4 a call | 1 | 14 |
| **THE BUBBLE**<br>a bubble gains on a man marching west; it stalls 12 px short of a man who stands still (measured) | **— (a buff in flight)**<br>flies 120 px/s at the front gnome; BIG on arrival | none on your men; it pops on a green round (1 HP) | 1 | 1 | 1 | 1 | one a fairy every 10 s | one a fairy every 10 s | one a fairy every 10 s | one a fairy every 10 s | 0 | 0 |
|  **THE WASP**<br>flies 78 px/s | **STING**<br>dives 520 px/s at a man within 260 px, stings every man his pass crosses, climbs 2.2 s between dives (half that while a hornet is on the floor) | **1 bar hit** + the man is **HELD 2 s** (no march, no gun) | 2 | 3 | 3 | 4 | 24 | 21 | 41 | 50 | 1 | 8 |

| unit | attack | what a hit does | HP EASY | HP MED | HP HARD | HP MEGA | n EASY | n MED | n HARD | n MEGA | leak | tags |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
|  **THE HORNET** HARD and MEGA · flies 60 px/s · while he lives every wasp dives twice as often | **HEAVY STING** hangs 0.4 s over his man (the tell) then dives 420 px/s at a man within 300 px, climbs 2.6 s | **3 bar hits** + the man is **bowled over** | — | — | 18 | 22 | — | — | 2 | 3 | 2 | 34 |
|  **THE POND FROG** born from the pool's rim; never walks | **SPLAT · TONGUE** hops 460 px/s at a man within 150 px (one a hop, every 2 s) · the tongue at a man within 56 px every 2 s · beside a man: a tongue and a hop about every 3 s each (measured: 6 bar hits in 7 s) | the splat **1 bar hit** + a shove · the tongue **1 bar hit** | 4 | 5 | 6 | 7 | 2 | 3 | 6 | 6 | 1 | 10 |

HP = each body as the palace's own wave rows bear it on that tier (measured in this read: the court ×1 · ×1.25 · ×1.5 · ×1.8, her bar ×2 · ×2.4 · ×6 · ×9); a BIG guard has four times his. n = bodies named in the solo wave sheet (one commander), THE COURT RETURNS rows included; her summons come on top ("+N a call"). leak = lives it costs at the toy box. tags = dog tags paid when it falls. Our rounds on the gnomes: a guard's shield turns every 2nd round from the front and bullets do half (blasts double); a knight's shield turns every round from the front (measured: 5 of 5 rifle rounds, his bar unmoved).

## Princess Croak, move by move

| move | EASY | MEDIUM | HARD | MEGA | how it works |
|---|---|---|---|---|---|
| **HARP SHOT** | **2 bar hits** · every 2.4 s | **2 bar hits** · every 2 s | **2 bar hits** · every 1.8 s | **3 bar hits** · every 2 s | The nearest man within 400 px (a jetting, cloaked or dragged man is not picked). Measured every 2.42 · 2.02 · 1.82 · 2.02 s, the first 1.6 s after she steps on; a level-70 rifleman takes 1.84 of his 7 bar hits a shot and falls to the 4th. She glides to 240 px from your line and backs off east at 96 px/s when a man closes inside 160 px. |
| **FACE SHOT** | **KO** · 0.47 s | **KO** · 0.47 s | **KO** · 0.47 s | **KO** · 0.47 s | A SARGE within 84 px (the Claymore, the pistol, any iron; never a man or a hero: measured). The 0.12 s crouch tell, the 80 px hop back (0.3 s), and the shot in the face: a KO sent with the saber's roll switched off (the KO door's rolled flag, by design: the auto-kill for a Sarge who comes too close), while THE RIOT SHIELD, Francis's block or the Claymore's counted block eat it whole and the deflect dome turns it. Measured: a saber Sarge at 60 px KO'd at 0.47 s on every tier; kept 60 px off her, the shotgun Sarge's plate ate 5 face shots and he fell to the 6th, the Claymore's block ate 1 and he fell to the 2nd; a rifleman at 60 px took the harp instead. |
| **LIGHT ARROWS** | off | **KO** · at 50 % of her bar | **KO** · at 50 % of her bar | **KO** · at 50 % of her bar and again at 25 % | MEDIUM and up: when her bar falls to half she roots and charges 2 s ("Let me show you true power."), then three bolts fly west from her, 120 px apart (her row and one above and below), 900 px/s: every man within 56 px of a bolt's line is KO'd through every save (a machine too); the bolts end at your deploy line, the toy box is never hit. Measured: three men in the three lanes KO'd 2 s after the gate, a medic 60 px off a lane untouched. MEGA fires it again at a quarter ("Punishment" in E Minor). On HARD and MEGA the arrow act also comes with her retreats (below). |
| **THE CATCH** | — | she takes **96 HP** (25 % of her bar) | she takes **240 HP** (25 % of her bar) | she takes **360 HP** (25 % of her bar) | A charged commando standing in a lane (or covering a man with THE BUDDY SHIELD), the blue wall, the teal's save or a swinging Claymore CATCHES a bolt: it flies back east at her, every gnome it crosses falls, and she takes a quarter of her bar (measured: the gnome in its path fell, her bar –25 %). A commando catches ONE bolt a fight (measured on MEGA: the second volley KO'd him). |

| move | EASY | MEDIUM | HARD | MEGA | how it works |
|---|---|---|---|---|---|
| **THE GATES** | 75 %: GUARDS! TO ME!<br>50 %: THE GUARD TURNS OUT<br>30 %: THE POND! EVERYONE OUT OF THE POND!<br>then THE COURT RETURNS ×5 every 40 s (2 gnome, 4 wasp) | 75 %: GUARDS! TO ME!<br>50 %: LET ME SHOW YOU TRUE POWER. — THE LIGHT ARROWS<br>30 %: THE POND! EVERYONE OUT OF THE POND!<br>then THE COURT RETURNS ×5 every 36 s (3 gnome, 3 wasp) | 75 %: GUARDS! TO ME!<br>50 %: LET ME SHOW YOU TRUE POWER. — THE LIGHT ARROWS<br>30 %: THE POND! EVERYONE OUT OF THE POND!<br>then THE COURT RETURNS ×8 every 24 s (6 gnome, 4 wasp) | 75 %: GUARDS! TO ME!<br>50 %: LET ME SHOW YOU TRUE POWER. — THE LIGHT ARROWS<br>30 %: THE POND! EVERYONE OUT OF THE POND!<br>25 %: AGAIN. — THE LIGHT ARROWS<br>then THE COURT RETURNS ×8 every 20 s (8 gnome, 5 wasp) | Her court arrives as her bar falls (each gate opens on the clock too: 56 · 48 · 42 · 42 s after it comes up). The gates latch once; a later bar does not reopen them. THE COURT RETURNS rows come on the clock alone, every 40 · 36 · 24 · 20 s. |
| **HER LIVES** | **1** bar | **1** bar | **3** bars of 960 | **3** bars of 1,440 | EASY and MEDIUM: her first zero is her fall. HARD and MEGA: THE FALL THAT ISN'T — at zero with bars left she flies back to her stage (≤ 1 s), holds 0.8 s, rises with a whole new bar and acts: HARD "summon" then "arrow", MEGA "both" twice. Measured on HARD: the retreat at 0.9 s, the court at 2.3 s; the second zero brought the arrows and KO'd the rifleman in her middle lane. Her third zero is her fall. |
| **THE BUBBLE COURT** | — | — | **3 guards + 3 fairies** | **4 guards + 4 fairies** | "MY FAIRIES! Get big my darlings!": the guards appear at her stage at ×1.5 / ×1.8 HP, each fairy on her own guard; the bubbles make them BIG (×4 HP) and the big guards CHARGE your line at ×2.2 speed (measured: 3 of 3 big within 4 s on HARD). Her fairies never mend. |
| **THE BAR CLOCK** | — | — | every **60 s** | every **45 s** | A later bar that stands 60 s (HARD) / 45 s (MEGA) calls THE BUBBLE COURT again (measured: the call at 60 s and 45 s). The fight always moves. |
| **THE WAVE AFTER HER FIRST BAR** | — | — | +3 guards + 3 fairies | +4 guards + 3 fairies | Once her first bar is down every COURT RETURNS wave also fields charging guards with a fairy each (the code's own rows; not staged on this page). |
| **WALK** | 24 px/s | 24 px/s | 24 px/s | 24 px/s | Her glide; 96 px/s on the back-off. A press she cannot gain a pixel on for 1.2 s is given up for a wander step (the throne room's columns and fountains). |
| **HER BAR (one commander)** | **320** | **384** | **960** | **1,440** | 160 HP × the tier's cell (×2 · ×2.4 · ×6 · ×9); HARD and MEGA three times over (lives). Her fall ends the fight. With more commanders the bar grows (the table below). |
| **THE TOY BOX** | –8 lives if she reaches it | –8 lives if she reaches it | –8 lives if she reaches it | –8 lives if she reaches it | Every other unit costs its LEAK (1–2) of the box's lives. |

### Her bar by the number of commanders

| | 1 commander | 2 commanders | 3 commanders | 4 commanders | 5 commanders | 10 commanders |
|---|---|---|---|---|---|---|
| **EASY** | 320 | 608 | 896 | 1,184 | 1,472 | 2,912 |
| **MEDIUM** | 384 | 730 | 1,075 | 1,421 | 1,766 | 3,494 |
| **HARD** | 960 | 1,824 | 2,688 | 3,552 | 4,416 | 8,736 |
| **MEGA** | 1,440 | 2,736 | 4,032 | 5,328 | 6,624 | 13,104 |

Read from the game's own wave sheet with 1, 2, 3, 4, 5 and 10 commanders in the fight. HARD and MEGA: three such bars (her lives). One Princess Croak at every seat count; her court scales with the party.

## What changes from EASY to MEGA

| | EASY | MEDIUM | HARD | MEGA |
|---|---|---|---|---|
| **Her harp shot (bar hits · seconds between)** | 2 · 2.4 | 2 · 2 | 2 · 1.8 | 3 · 2 |
| **Her light arrows** | — | at 50 % | at 50 % | **at 50 % and 25 %** |
| **Her lives (bars)** | 1 | 1 | 3 | 3 |
| **Her bar (one commander)** | 320 (×2) | 384 (×2.4) | 960 (×6) | 1,440 (×9) |
| **The court's HP** | ×1 | ×1.25 | ×1.5 | ×1.8 |
| **Guards · slingers · knights · sages in the sheet** | 20 · 2 · — · — | 27 · 3 · 2 · 1 | 72 · 6 · 4 · 3 | 94 · 7 · 5 · 5 |
| **Wasps · hornets · frogs · fairies in the sheet** | 24 · — · 2 · 1 | 21 · — · 3 · 2 | 41 · 2 · 6 · 2 | 50 · 3 · 6 · 2 |
| **Bodies in the sheet, solo (she is not counted)** | 49 | 59 | 136 | 172 |
| **Gate clock (a gated wave opens by itself after)** | 56 s | 48 s | 42 s | 42 s |
| **THE COURT RETURNS (waves × every)** | 5 × 40 s | 5 × 36 s | 8 × 24 s | 8 × 20 s |
| **THE BUBBLE COURT at a retreat (guards + fairies) · the bar clock** | — | — | 3 + 3 · 60 s | 4 + 4 · 45 s |
| **The room** | day | day | day | **moonlit** (night) |
| **The lab's stock army level** | 70 | 77 | 83 | 90 |

## The wave sheets (solo, one commander)

### EASY · 49 bodies · level 70

1. **PRINCESS CROAK TAKES THE STAIR** (on the clock)
   4 gnome guard, 2 slinger, 1 fairy · PRINCESS CROAK (bar 320)
2. **GUARDS! TO ME!** (holds while princess stands above 75% (or 56 s))
   2 wasp, 1 pond frog
3. **THE GUARD TURNS OUT** (holds while princess stands above 50% (or 56 s))
   3 gnome guard, 2 wasp
4. **THE POND! EVERYONE OUT OF THE POND!** (holds while princess stands above 30% (or 56 s))
   1 pond frog, 3 gnome guard
5. **GUARDS! TO ME!** (holds while princess stands above 0% (or 40 s))
   2 gnome guard, 4 wasp × 5

### MEDIUM · 59 bodies · level 77

1. **PRINCESS CROAK TAKES THE STAIR** (on the clock)
   5 gnome guard, 3 slinger, 1 fairy · PRINCESS CROAK (bar 384)
2. **GUARDS! TO ME!** (holds while princess stands above 75% (or 48 s))
   2 knight, 1 fairy, 4 wasp, 2 pond frog
3. **LET ME SHOW YOU TRUE POWER.** (holds while princess stands above 50% (or 48 s))
   4 gnome guard, 2 wasp
4. **THE POND! EVERYONE OUT OF THE POND!** (holds while princess stands above 30% (or 48 s))
   1 pond frog, 3 gnome guard, 1 sage
5. **GUARDS! TO ME!** (holds while princess stands above 0% (or 36 s))
   3 gnome guard, 3 wasp × 5

### HARD · 136 bodies · level 83

1. **PRINCESS CROAK TAKES THE STAIR** (on the clock)
   10 gnome guard, 6 slinger, 1 fairy · PRINCESS CROAK (bar 960)
2. **GUARDS! TO ME!** (holds while princess stands above 75% (or 42 s))
   4 knight, 1 fairy, 7 wasp, 2 hornet, 3 pond frog
3. **LET ME SHOW YOU TRUE POWER.** (holds while princess stands above 50% (or 42 s))
   8 gnome guard, 2 wasp
4. **THE POND! EVERYONE OUT OF THE POND!** (holds while princess stands above 30% (or 42 s))
   3 pond frog, 6 gnome guard, 3 sage
5. **GUARDS! TO ME!** (holds while princess stands above 0% (or 24 s))
   6 gnome guard, 4 wasp × 8

### MEGA · 172 bodies · level 90

1. **PRINCESS CROAK TAKES THE STAIR** (on the clock)
   12 gnome guard, 7 slinger, 1 fairy, 3 sage · PRINCESS CROAK (bar 1,440)
2. **GUARDS! TO ME!** (holds while princess stands above 75% (or 42 s))
   5 knight, 1 fairy, 8 wasp, 3 hornet, 3 pond frog
3. **LET ME SHOW YOU TRUE POWER.** (holds while princess stands above 50% (or 42 s))
   10 gnome guard, 2 wasp
4. **THE POND! EVERYONE OUT OF THE POND!** (holds while princess stands above 30% (or 42 s))
   3 pond frog, 8 gnome guard, 2 sage
5. **AGAIN.** (holds while princess stands above 25% (or 42 s))
   nothing new: her second volley

6. **GUARDS! TO ME!** (holds while princess stands above 0% (or 20 s))
   8 gnome guard, 5 wasp × 8

She opens on her top step; her court walks on behind her over the first seconds. Each gated wave opens when her bar falls to its mark, or by itself after the gate clock. THE COURT RETURNS rows come on the clock alone, whatever her bar.

## How this was read and checked

The live client v28.88 (index.html md5 1de0910be46cda954034c14925273a12, the bytes deployed on the Big Mac as v28.88) was opened headless on this Mac and read with the game's own functions: every table, every tier's wave sheet, a real lab fight per tier for the stock army and its level, every man's HP and bar at every rank, her bar for 1 to 10 commanders. Then each hit was **measured on the real machines** in a BOSS LAB fight on every tier, every attacker born through the palace's own wave row so the tier's stamps ride: her harp shot on all eleven men, Sarge's five irons and the six heroes; her face shot on a saber and a pistol Sarge (and that a rifleman, a hero and a Sarge at 100 px are spared); her light arrows on three men in the three lanes and a medic between them, the commando's catch and the ray thrown back at her and her gnome, MEGA's second volley; the guard's jab and ram (and the scout jeep, the V2 jeep and the tank on the spear), the knight's jab and shield, the guard's shield and armour under our rifle, the slinger's glob, the sage's toad curse and fire, the fairy's mend and bubble, the wasp's sting and the hold, the hornet's sting and the bowl-over, the frog's splat and tongue; on HARD and MEGA her retreat, THE BUBBLE COURT, the big guards' charge, the bar clock and the second zero; every body as the tier bears it. Chrome 244/244 · WebKit 244/244 checks passed, zero page errors. The two engines' READS are identical (checked), and every damage fact in the two proofs agrees (1292 facts checked across the engines: each hit's weight, every KO and every save, every body). What differs between Chrome and WebKit is dice and clocks, and this page prints both where it matters: the saber Sarge's harp shots landed to his KO (9/11 · 12/9 · 7/7 · 5/7 on E · M · H · G); the gnome's ram on 16 fresh saber Sarges (6/10 · 7/8 · 6/3 · 9/11 KO'd); EMPEROR BEAR's shots landed in 60 s (EASY 20/22, MEDIUM 23/25, MEGA 21/16); PRINCESS CROAK's shots landed in 60 s (EASY 20/21, MEDIUM 25/24); the frame a hit lands on, and the slinger's wander.

Sources (the code's own names): FDEF (princess · gnome · gslinger · gsage · gknight · fairy · fbubble · wasp · hornet · pfrog); updateFoePrincess (PC_SHOT, the halt and back-off, pcNearest, pcFaceShot → knock(rolled), the gates → pcFire), the 'lightarrow' shot branch (knockKill, cmdoCharged/cmdoCovers, tkDeflT, mblRay, arrowBack → hurt under TK_REFL), PC_ARROW_TIER, PC_LIVES, pcRetreat / pcRetTick (PC_RET, PC_RET_ACTS), pcSummon / pcSummonIn (PC_SUMMON, PC_COURT_HPX), PC_BAR_CLK, pcwWave (PCW), PC_GIANT_SPDX; updateFoeTrike (the ram bag → knock; the vehicle branch → knockKill), gnJabTick, gnSlingTick + tanShoot (def.pips → pipN), gnCasterTick (fsIgnite → fsTick: tanHit fire beats; toadT), gnFairyTick / gnBubbleTick (big, bigX), updateFoeSwoop (sting.pips → tanHit, kyHeldT, ggShove), updateFoePounce (pounce.pips, tongue), hurt (the gnomes' shield shN, gnArmorX); tanHit, DSM (k 0.9), dsmH / dsmX, dhtH, gritCap, mgCtr, enhSoakM, saberBlocks, sargeAbsorbs (ENH_SHIELD_N), cmdAbsorbs; raidCfg's palace branch (PC_HPX, the court's cells, hpGate rows with clk, PC_GATE_CLK, PC_RETURN_CLK), blbSheet('palace', tier, N), blbStockDef (BLB_STOCK_LV.palace), blbLaunch; bfHpAt, usBfHP, sgBfHP, heroBfHP. Kit: design/boss-princess-unit-breakdown-1031/ (kit/readPalace.mjs · kit/provePalace.mjs · kit/build.py · kit/render.mjs · kit/export_dash.py; rig/lrbShared.mjs; data/; proof/logs/). Zero game bytes.